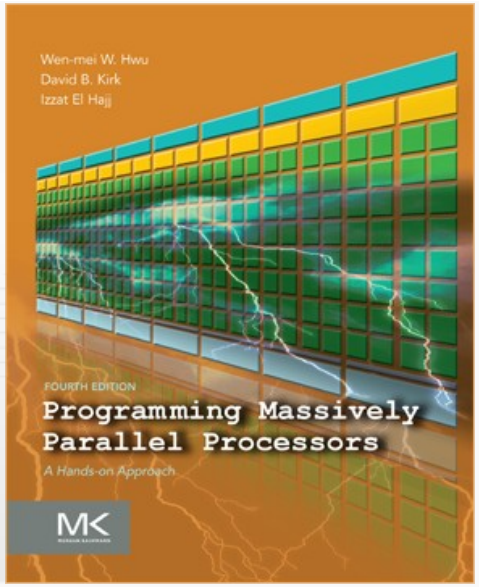
Programming Massively Parallel Processors

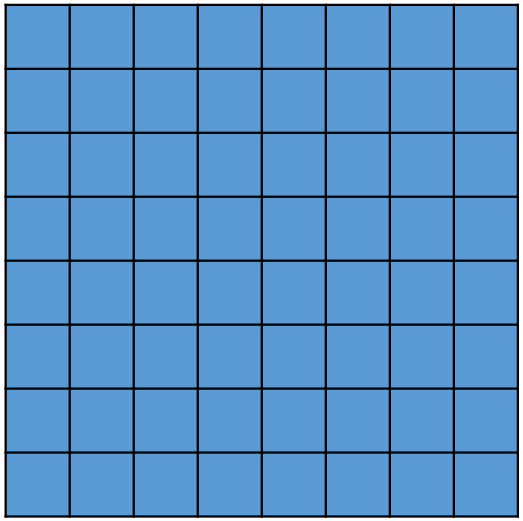
A Hands-on Approach

CHAPTER 14 (PART 1) **Sparse Matrix Computation**



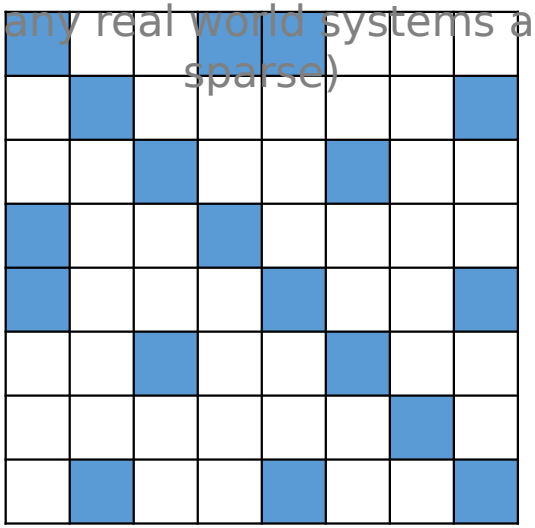
Sparse Matrices

A **dense matrix** is one where the majority of elements are not zero



A **sparse matrix** is one where many elements are zero

(many real world systems are sparse)



- Opportunities:
  - Do not need to allocate space for zeros (save memory capacity)
  - Do not need to load zeros (save memory bandwidth)
  - Do not need to compute with zeros (save computation time)

Sparse Matrix Storage Formats

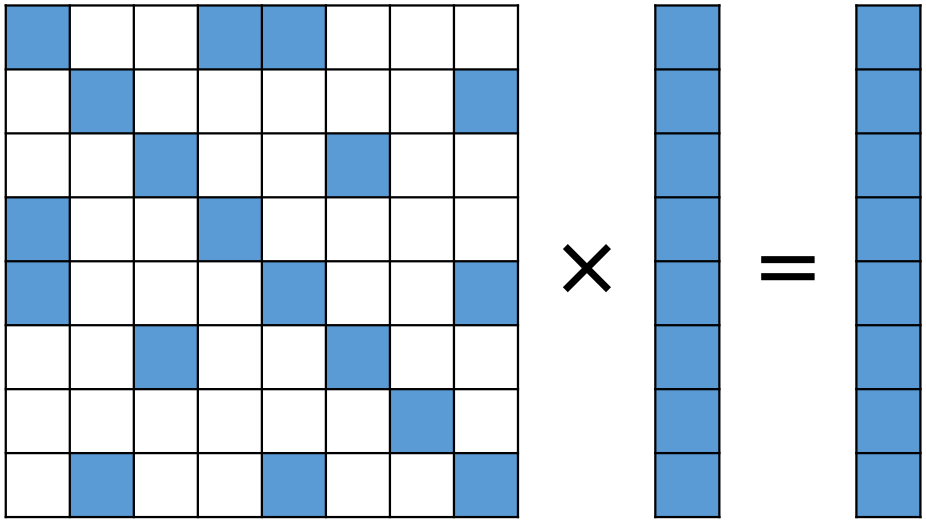
- Many storage formats for sparse matrices:
  - Coordinate Format (COO)
  - Compressed Sparse Row (CSR)
  - ELLPACK Format (ELL)
  - Jagged Diagonal Storage (JDS)
  - …

- Format design considerations:
  - Space efficiency (memory consumed)
  - Flexibility (ease of adding/reordering elements)
  - Accessibility (ease of finding desired data)
  - Memory access pattern (enabling coalescing)
  - Load balance (minimizing control divergence)

SpMV

- Choice of best format depends on computation

- We will use **Sparse Matrix-Vector multiplication** (SpMV) as an example to study different formats



Coordinate Format (COO)

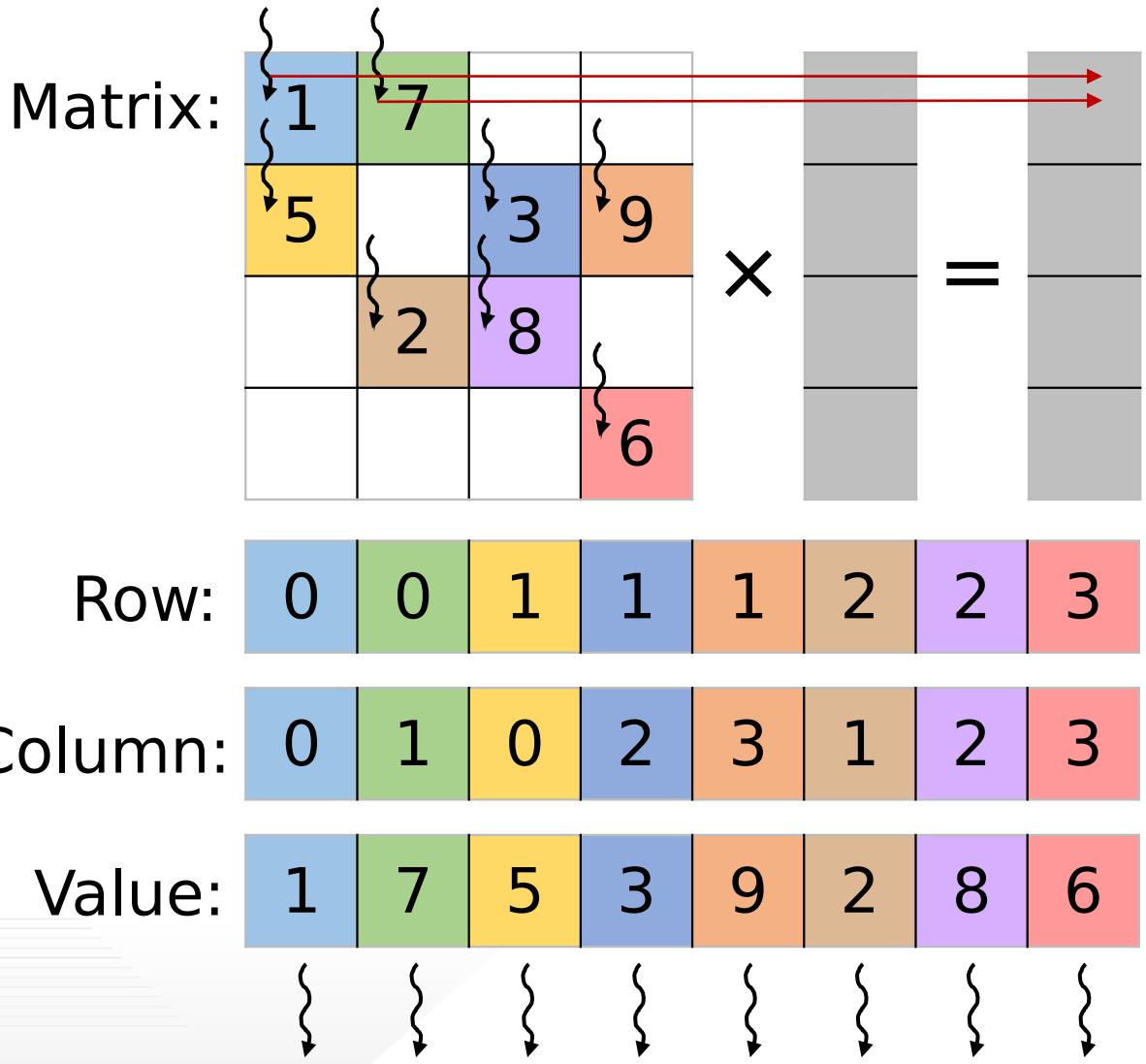
Matrix:

1	7		
5		3	9
	2	8	
			6

Store every nonzero along with its row index and column index

Row:	0	0	1	1	1	2	2	3
Column:	0	1	0	2	3	1	2	3
Value:	1	7	5	3	9	2	8	6

SpMV/COO



Multiple threads writing to the same output (need atomic operations)

Parallelization approach:
Assign one thread per nonzero

SpMV/COO Code

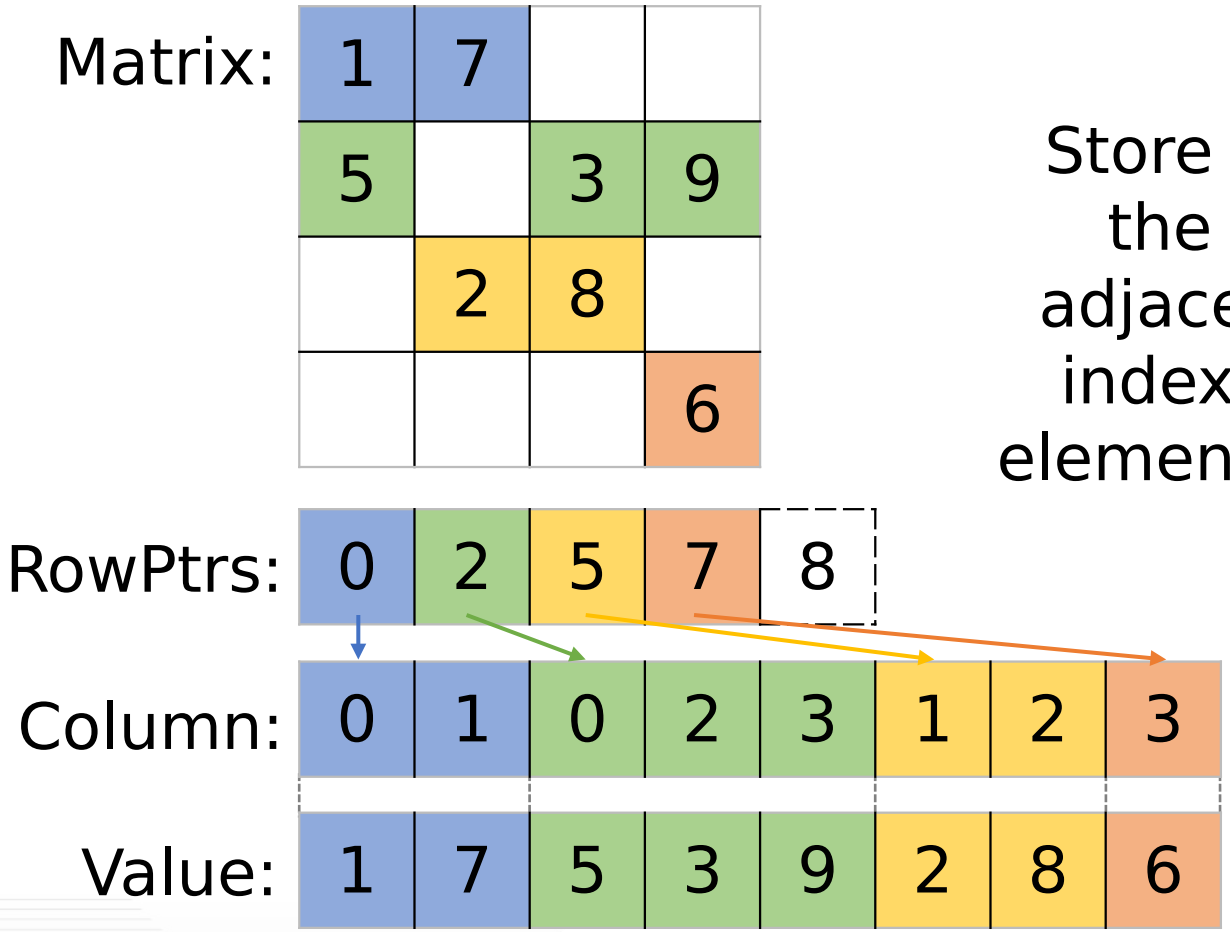
```
__global__ void spmv_coo_kernel(COOMatrix cooMatrix, float* inVector, float* outVector) {
    unsigned int i = blockIdx.x*blockDim.x + threadIdx.x;
    if(i < cooMatrix.numNonzeros) {
        unsigned int row = cooMatrix.rowIdxs[i];
        unsigned int col = cooMatrix.colIdxs[i];
        float value = cooMatrix.values[i];
        atomicAdd(&outVector[row], inVector[col]*value);
    }
}
```

COO Tradeoffs

- Advantages:
  - Flexibility: easy to add new elements to the matrix, nonzeros can be stored in any order
  - Accessibility: given nonzero, easy to find row and column
  - SpMV/COO has coalesced memory accesses
  - SpMV/COO has no control divergence

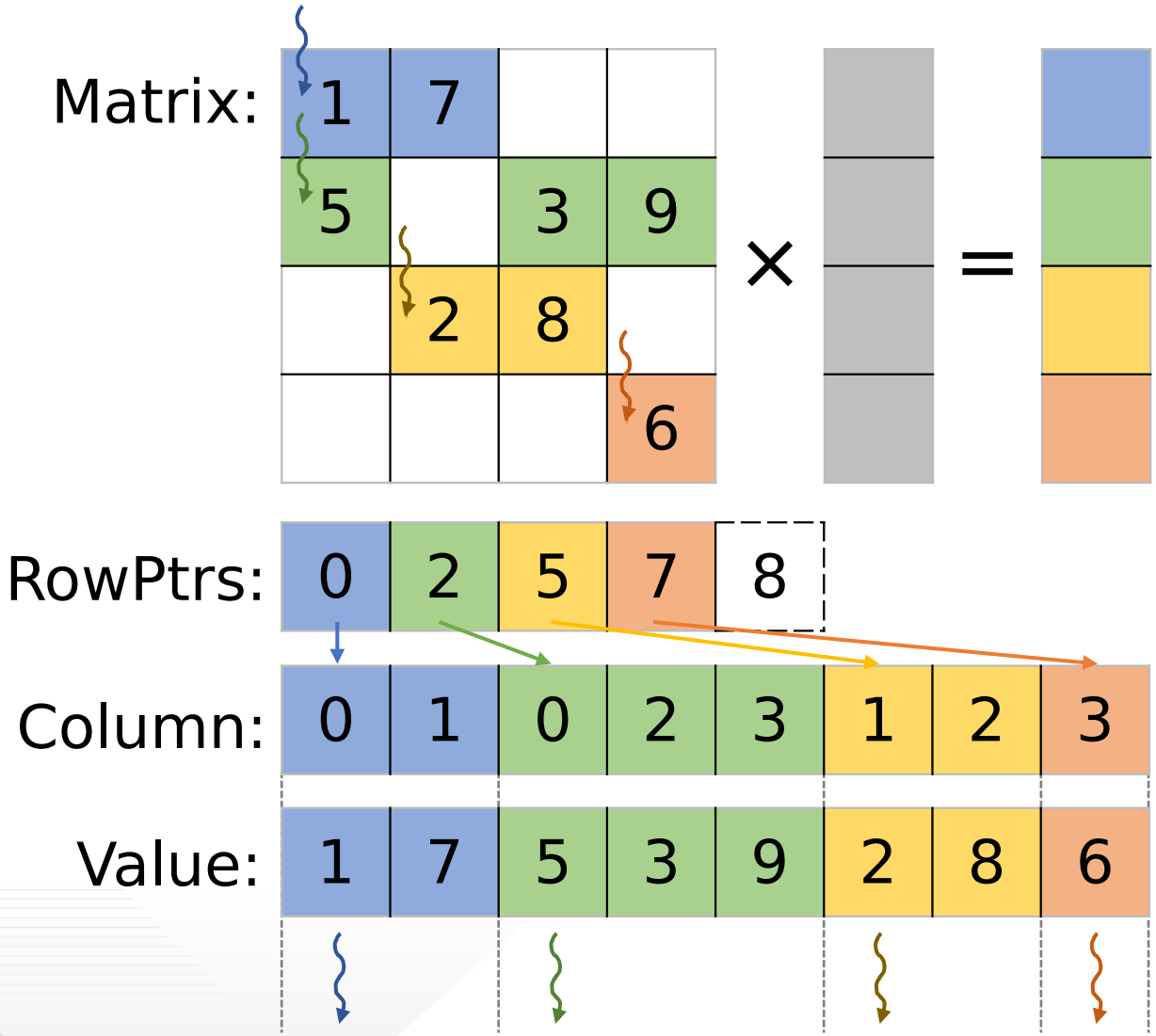
- Disadvantage:
  - Accessibility: given a row or column, hard to find all nonzeros (need to search)
  - SpMV/COO uses atomic operations

Compressed Sparse Row (CSR)



Store nonzeros of the same row adjacently and an index to the first element of each row

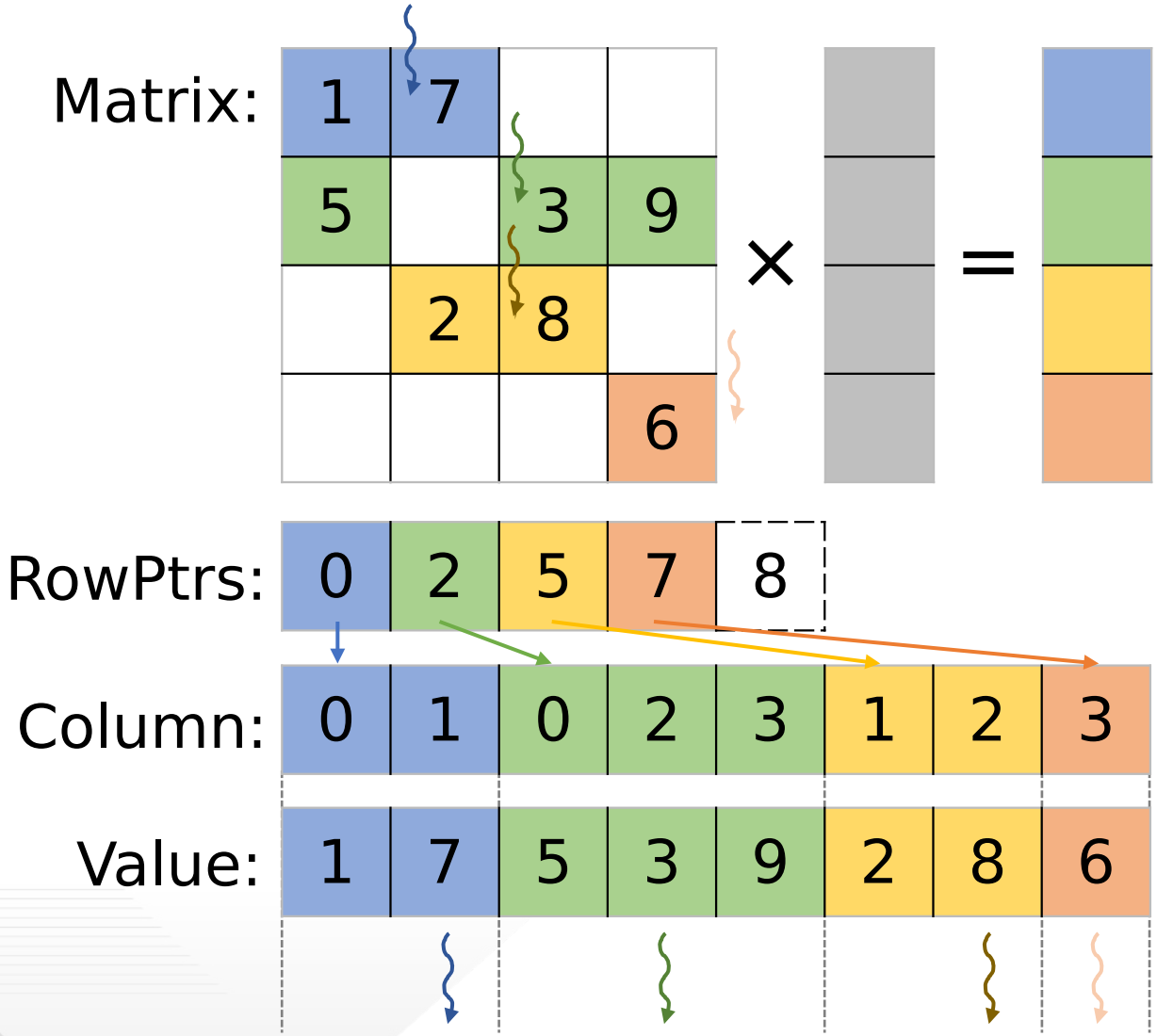
SpMV/CSR



Parallelization approach:

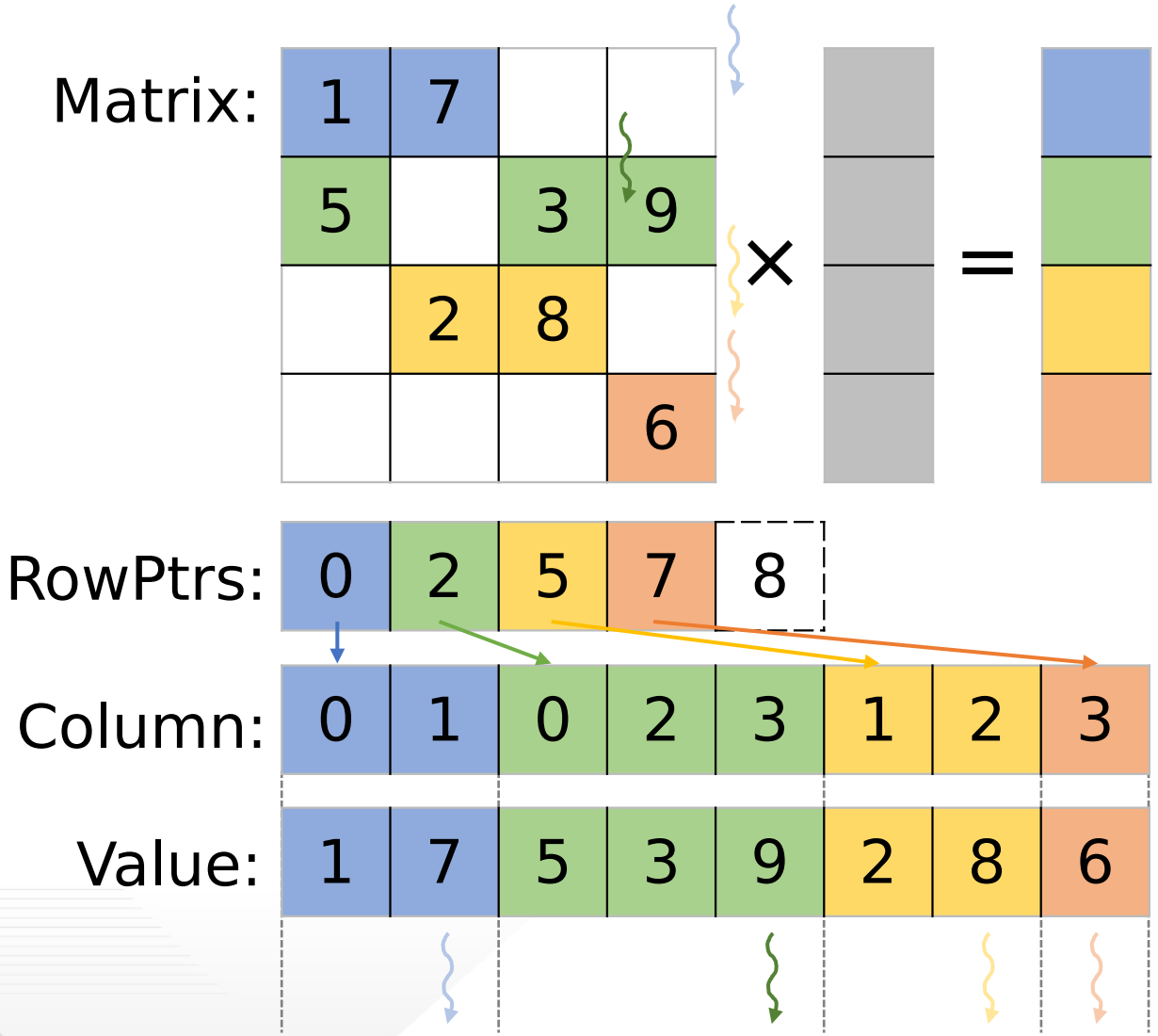
Assign one thread to loop over each input row sequentially and update corresponding output element

SpMV/CSR



Parallelization approach:

Assign one thread to loop over each input row sequentially and update corresponding output element



Parallelization approach:

Assign one thread to loop over each input row sequentially and update corresponding output element

SpMV/CSR Code

```
__global__ void spmv_csr_kernel(CSRMatrix csrMatrix, float* inVector, float* outVector) {
    unsigned int row = blockIdx.x*blockDim.x + threadIdx.x;
    if(row < csrMatrix.numRows) {
        float sum = 0.0f;
        for(unsigned int i = csrMatrix.rowPtrs[row]; i < csrMatrix.rowPtrs[row + 1]; ++i) {
            unsigned int col = csrMatrix.colIdxs[i];
            float value = csrMatrix.values[i];
            sum += inVector[col]*value;
        }
        outVector[row] = sum;
    }
}
```

CSR Tradeoffs (versus COO)

- Advantages:
  - Space efficiency: row pointers smaller than row indexes
  - Accessibility: given a row, easy to find all nonzeros
  - SpMV/CSR avoids atomics, every thread owns its output

- Disadvantage:
  - Flexibility: hard to add new elements to the matrix
  - Accessibility: given nonzero, hard to find row; given a column, hard to find all nonzeros
  - SpMV/CSR memory accesses are not coalesced
  - SpMV/CSR has control divergence

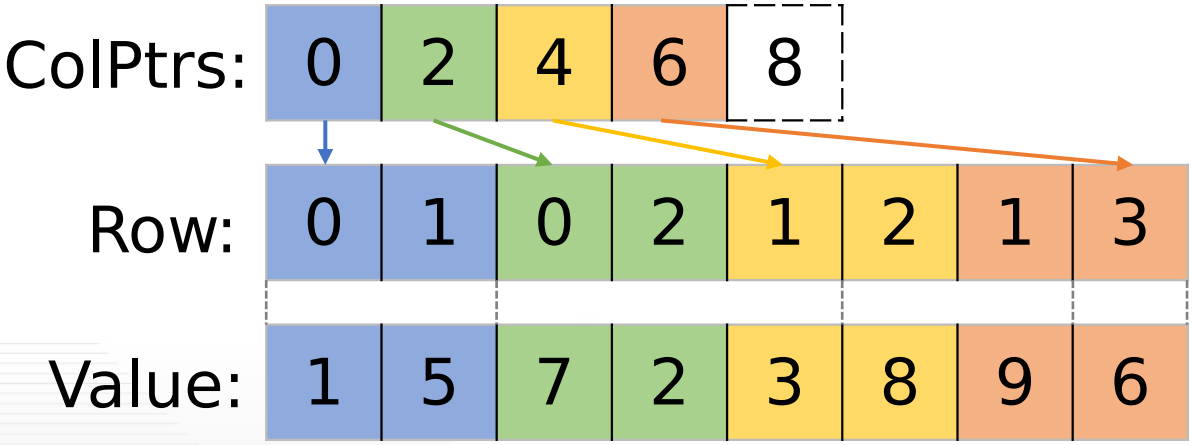
Compressed Sparse Column (CSC)

Matrix:

1	7		
5		3	9
	2	8	
			6

Like CSR, but groups nonzeros by column

(useful for computations other than SpMV that require column traversal, such as SpMSpV)



References

- Wen-mei W. Hwu, David B. Kirk, and Izzat El Hajj. *Programming Massively Parallel Processors: A Hands-on Approach*. Morgan Kaufmann, 2022.